Implied Correlation Arbitrage: A Dispersion-Based Options Strategy for Equity ETFs

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Group 5: Pascal Bermeo Neumann, Sarang Deshpande, Ethan Fine, Michael Waltuch

1 Introduction

Options markets embed forward-looking information about expected volatility and correlation that is not available from historical price data alone. When a portfolio of assets is traded as a single instrument, as in the case of an exchange-traded fund (ETF), its options market generates an implied volatility that reflects the market's aggregate expectation about the basket's future variance. Separately, options on each individual constituent encode expectations about that constituent's own volatility. Under the assumption of correlated geometric Brownian motion, these two sets of information are linked by a no-arbitrage constraint: the ETF's implied variance must equal the weighted sum of constituent implied variances scaled by their pairwise correlations. When the ETF's market-quoted implied volatility persistently deviates from the volatility implied by its constituents, a dispersion trading opportunity arises.

Dispersion trading exploits this wedge by selling ETF options and simultaneously buying options on the underlying constituents. The core intuition is that ETF options tend to embed a premium for correlation risk, since institutional investors frequently use ETF options to hedge broad portfolio exposures. This demand-driven premium pushes implied correlation, derived from the ETF implied variance and the constituent implied variances, above realized correlation on average. The strategy monetizes this premium by going short ETF variance and long constituent variance, with consistent notional sizing across ETFs. Profits arise when implied correlation reverts toward its realized level.

The primary signal in this strategy is the *implied correlation spread*, defined as the difference between the ETF's market-quoted implied volatility and the synthetic basket implied volatility derived from constituent options. The basket implied volatility is computed using the standard portfolio variance formula under correlated GBM:

$$\sigma^2_{\text{basket}} = \sum_i \sum_j w_i \, w_j \, \sigma_i \, \sigma_j \, \rho_{ij} \tag{1}$$

where w_i are constituent weights from ETF holdings snapshots, σ_i are constituent at-the-money implied volatilities from OptionMetrics matched to the relevant expiry, and ρ_{ij} is the pairwise correlation matrix estimated from CRSP daily returns using a 60-day rolling Ledoit-Wolf shrinkage estimator. A persistently positive spread indicates that the ETF options market is overpricing correlation relative to what the constituent options surface implies. We enter the trade when the spread exceeds one standard deviation above its rolling 60-day mean and exit when it reverts to the mean.

While dispersion trading has been studied extensively in the context of large, liquid indices such as the S&P 500, less attention has been paid to how the strategy performs across ETFs of varying liquidity. This distinction matters for several reasons. Less liquid ETFs tend to have wider bid-ask spreads and thinner options markets, which can allow mispricings to persist longer before being arbitraged away. At the same time, transaction costs are higher in these markets, which may erode or eliminate the theoretical edge. The interaction between mispricing persistence and transaction cost drag creates a natural question: is there an intermediate liquidity tier at which dispersion trading is most profitable?

This paper investigates that question by applying the dispersion strategy to a cross-section of equity ETFs spanning three liquidity tiers. The high-liquidity tier, represented by SPY, serves as a negative control where we expect the implied correlation spread to be narrow and short-lived due to active arbitrage. The medium-liquidity tier, represented by XLK, and the low-liquidity tier, represented by XBI, are where we expect to find more persistent mispricings. Our central empirical hypothesis is that the implied correlation spread is wider and more persistent in less liquid ETFs, and that net strategy returns increase with decreasing liquidity up to the point where transaction costs dominate.

We test this hypothesis using data from OptionMetrics and CRSP spanning January 2022 through December 2024, a period that captures both a high-volatility regime driven by macroeconomic uncertainty and a subsequent period of volatility compression. This range allows us to assess strategy robustness across meaningfully different market conditions. All option filtering uses log-moneyness within a symmetric ± 0.15 band to ensure consistency across ETFs with different absolute price levels.

2 Data

2.1 Sources

This project draws on three primary data sources: OptionMetrics (accessed via the Wharton Research Data Services platform), CRSP, and constituent holdings files obtained directly from ETF providers.

OptionMetrics is the source for all options-related data. Daily option prices, implied volatilities, and Greeks are pulled from the `opprcd` annual tables (`opprcd2022`, `opprcd2023`, `opprcd2024`), which are unioned across years to cover the full sample period. Implied volatilities in OptionMetrics are com-

puted using a binomial tree model that accounts for early exercise and discrete dividends, ensuring internal consistency across strikes and maturities. The risk-free rate used in those calculations is the OptionMetrics zero curve, available in `optionm.zerocd`, and we use the same curve in our own Black-Scholes calculations to maintain consistency. ETF spot prices are sourced from `optionm.uprd`, the underlying price table for individual securities and ETFs. Security identifiers (`secid`) are resolved from tickers using `optionm.securd`, filtering to `issue_description = 'Intraday'` to select the correct entry for ETFs that have multiple rows in the table.

CRSP provides the daily constituent return data used to estimate the pairwise correlation matrix. Daily log returns are computed from closing prices in `crsp.dsf` and matched to OptionMetrics via a ticker-to-PERMNO crosswalk through `crsp.stocknames`.

ETF holdings are sourced as CSV snapshots from the SPDR provider websites. We collect one snapshot per calendar year (2022, 2023, 2024) per ETF and treat each snapshot as static within that calendar year. Constituent turnover within a given year is modest for the technology sector ETFs in our sample, while the biotechnology sector ETF saw more turnover in securities, with only 55 percent of securities remaining in the ETF across all snapshots. This simplification avoids the complexity of daily rebalancing at the cost of a small approximation error at year boundaries.

2.2 Sample Period and ETF Universe

As mentioned in the introduction, the sample covers January 1, 2022 through December 31, 2024. This window is chosen deliberately to span two meaningfully different volatility regimes: 2022 was characterized by elevated and rising volatility driven by the Federal Reserve tightening cycle and broad equity drawdowns, while 2023 and 2024 exhibited sustained volatility compression alongside a strong equity recovery. Testing the dispersion strategy across both regimes allows us to assess whether the implied correlation spread and its associated trading signal are regime-dependent.

The ETF universe spans three liquidity tiers. SPY serves as the high-liquidity negative control, where active arbitrage is expected to keep the implied correlation spread narrow. XLK represents the medium-liquidity tier, with a concentrated holdings structure of approximately 25 names and highly liquid options on both the ETF and its top constituents. XBI represents the low-liquidity tier, where thinner options markets may allow mispricings to persist for longer. For each ETF, constituent coverage is capped at the top N names by portfolio weight: 50 for SPY, 25 for XLK, and 20 for XBI. The caps keep the correlation matrix dimension well below the rolling window length, which is the regime in which the shrinkage estimator is most effective.

2.3 Filtering and Processing

Options are filtered at the query level before any downstream processing. The days-to-expiry window is restricted to 30 to 60 days to focus on the near-term volatility surface where trading activity is concentrated and implied volatility estimates are most reliable. Minimum implied volatility is set at 1% and maximum at 50% to exclude stale quotes and deep-in-the-money options with unreliable IV calculations.

Moneyness is parameterized throughout using log-moneyness, defined as $m = \ln(K/S)$, where K is the strike price and S is the spot price. Options are retained if $|m| \leq 0.15$, forming a symmetric band around the at-the-money point. Raw moneyness with a one-sided cap is not used because it asymmetrically drops out-of-the-money calls while retaining puts at the same relative distance from spot, which would bias the implied volatility surface. The log-moneyness parameterization also has the desirable property of being symmetric: a call and a put equidistant from the money in log-moneyness space are treated equivalently.

Bid-ask spread quality is enforced by computing the percentage spread, defined as $(\text{ask} - \text{bid})/\text{mid}$, and excluding options where this exceeds 10%. This threshold removes illiquid options with quotes that would impose excessive transaction costs without providing meaningful information about the implied volatility surface.

The primary signal is constructed from at-the-money options, defined here as options with $|m| < 0.025$. The full filtered surface, covering the entire ± 0.15 log-moneyness band, is retained separately for volatility surface visualization. Mid prices are computed as $(\text{bid} + \text{ask})/2$, and time to maturity in years is computed as $(\text{expiry} - \text{date})/365$.

Row count diagnostics are recorded after every data pull and after each filtering step, grouped by security, to ensure that no security is silently dropped by a filter that happens to be misconfigured for a particular ETF's price level or options structure.

3 Methodology

3.1 Overview

The methodology proceeds in five sequential stages: (1) basket variance computation, (2) implied correlation spread construction, (3) signal generation, (4) strategy execution and delta hedging, and (5) performance evaluation. Each stage maps to a distinct module in the codebase, with clean interfaces between them so that each component can be tested independently. The model layer (`basket_vol.py` and `correlation.py`) is purely mathematical and carries no knowledge of trading logic. The signal layer (`signal.py`) consumes model outputs and produces actionable entry and exit flags without touching position management. The strategy layer (`strategy.py` and `backtest.py`) turns those flags into positions and computes realized performance. This separation keeps

each module narrowly responsible and makes it straightforward to substitute components, for example swapping in a different correlation estimator, without cascading changes across the pipeline.

3.2 Correlation Matrix Estimation

The pairwise correlation matrix used in the basket variance formula is estimated from CRSP daily log returns using a 60-day rolling window. At each date t, the return submatrix $\mathbf{R}_t$ of shape $(60 \times N)$ is extracted for the selected constituents and mean-centered. A Ledoit-Wolf shrinkage estimator is then applied to produce a regularized covariance matrix:

$$\hat{\Sigma}_{\mathrm{LW}} = (1-\alpha)\,\mathbf{S} + \alpha\,\mu\,\mathbf{I} \tag{2}$$

where $\mathbf{S}$ is the sample covariance matrix, α is the optimal shrinkage intensity solved analytically by minimizing the expected Frobenius distance between the shrunk estimate and the true covariance, μ is the mean diagonal element of $\mathbf{S}$, and $\mathbf{I}$ is the identity matrix. The shrinkage target is the scaled identity, which pulls the off-diagonal correlations toward zero and the variances toward a common mean. This is appropriate in a setting where we have no strong prior about the direction of the estimation error in any particular pairwise correlation.

The correlation matrix is then extracted from the regularized covariance by normalizing by the diagonal standard deviations:

$$\hat{\rho}_{ij} = \frac{\hat{\Sigma}_{\mathrm{LW},ij}}{\sqrt{\hat{\Sigma}_{\mathrm{LW},ii}\,\hat{\Sigma}_{\mathrm{LW},jj}}} \tag{3}$$

Any constituent with a missing return in the rolling window is dropped from the matrix for that date. The surviving tickers are re-intersected with the holdings snapshot weights and renormalized before computing the basket variance, ensuring the weights always sum to one.

The average shrinkage intensity across the sample period was 0.19 for SPY, 0.14 for XLK, and 0.48 for XBI. The substantially higher shrinkage for XBI reflects the smaller constituent count (20 names) relative to the 60-day estimation window, which produces a noisier sample covariance. Heavy shrinkage pulls off-diagonal correlations toward zero, mechanically lowering the estimated basket variance and widening the implied correlation spread. The average pairwise correlation after shrinkage was 0.22 for SPY, 0.39 for XLK, and 0.12 for XBI. This suggests that a meaningful portion of XBI's large spread level is attributable to estimation bias rather than a genuine market mispricing.

3.3 Basket Variance and Synthetic Implied Volatility

The synthetic ETF implied volatility is computed in `src/model/basket_vol.py` using the portfolio variance formula under correlated GBM:

$$\sigma^2_{\mathrm{basket}} = \mathbf{v}^\top \hat{\boldsymbol{\rho}}\,\mathbf{v} \tag{4}$$

where $\mathbf{v}$ is the vector of constituent volatility-weight products, with elements $v_i = w_i \sigma_i$, and $\hat{\boldsymbol{\rho}}$ is the Ledoit-Wolf shrinkage correlation matrix estimated from CRSP returns. This is implemented directly as the matrix product `weighted_vols @ corr_matrix @ weighted_vols`.

The construction uses a hybrid approach for the two inputs. The diagonal volatility terms σ_i are sourced from constituent ATM implied volatilities in OptionMetrics, selecting the option with the smallest $|m|$ within the 30-to-60-day expiry window on each date. The off-diagonal correlation structure $\hat{\rho}_{ij}$ is the Ledoit-Wolf realized estimate from CRSP returns described in Section 3.2. This hybrid is intentional: options markets on individual constituents directly reveal the market's expectation of each name's own future variance, while realized correlations provide the most tractable and stable estimate of cross-constituent dependence given the varying liquidity of constituent options across the three ETF tiers.

On each date t, the calculator intersects three sets: the constituents present in the holdings snapshot, the tickers for which the correlation estimator has a valid diagonal entry, and the tickers for which a non-missing ATM implied volatility is available. Only constituents in all three sets are included. If fewer than two constituents survive the intersection on a given date, σ_{basket} is recorded as `NaN` and that date is excluded from the signal computation. Weights are renormalized over the surviving constituent set so they sum to one, ensuring the basket variance remains well-defined under partial constituent coverage. Constituent implied volatilities that are missing on isolated dates are forward-filled before the intersection check, so that a single day of missing data for a constituent does not unnecessarily drop that name from the basket.

3.4 Implied Correlation Spread and Signal Generation

Signal generation is implemented in `src/signal/signal.py`. The implied correlation spread on date t for ETF e is:

$$\Delta\sigma_e(t) = \sigma_{\text{ETF},e}(t) - \sigma_{\text{basket},e}(t) \tag{5}$$

where $\sigma_{\text{ETF},e}(t)$ is the market-quoted ATM implied volatility of the ETF, selected as the option with the smallest $|\ln(K/S)|$ on date t within the 30-to-60-day expiry window.

The raw spread is standardized into a z-score using a rolling window of length $W = 60$ trading days:

$$z_e(t) = \frac{\Delta\sigma_e(t) - \bar{\Delta\sigma}_e(t)}{s_{\Delta\sigma,e}(t)} \tag{6}$$

where $\bar{\Delta\sigma}_e(t)$ and $s_{\Delta\sigma,e}(t)$ are the rolling mean and standard deviation of the spread over the preceding 60 trading days. Using a z-score rather than a raw spread threshold normalizes for the large level differences across ETF tiers visible in Figure 1: XBI's spread is structurally 20 to 30 volatility points above zero for most of the sample, while SPY and XLK oscillate near zero, so a single raw

threshold would be permanently triggered for XBI and never triggered for SPY. The z-score transformation places all three ETFs on a common scale.

Entry and exit conditions are:

$$\text{Entry:} \quad z_e(t) > 1.0 \tag{7}$$

$$\text{Exit:} \quad z_e(t) < 0.0 \tag{8}$$

Entry is triggered when the spread is more than one standard deviation above its rolling mean, indicating that the ETF options market is pricing a wider correlation premium than usual relative to recent history. Exit is triggered when the z-score returns to zero, indicating the spread has reverted to its rolling mean. These thresholds were chosen to balance signal frequency against the transaction cost budget: a higher entry threshold produces fewer, cleaner trades but reduces gross PnL, while a lower threshold generates more trades at higher aggregate cost. At $z = 1.0$, the strategy produced between 9 and 17 trades per ETF over the three-year sample, as shown in Table 1.

3.5 Strategy Execution

The strategy is implemented in `src/strategy/strategy.py` and operates on a consistent notional of \$100,000 per trade across all three ETFs, ensuring that cross-ETF performance comparisons are not confounded by differences in position size.

On each entry date, a short straddle is opened on the ETF by selling the ATM call and ATM put closest to $|m| = 0$ within the available option chain. The short straddle position profits when the ETF's implied volatility subsequently declines, which is the expected outcome when the implied correlation spread reverts. Simultaneously, long straddle positions are opened on constituent names, notionally weighted by w_i, representing the long variance leg of the dispersion trade. Both legs are marked to market daily. The constituent long straddles are weighted by w_i and their mid-price changes contribute to daily PnL alongside the short ETF straddle.

Daily PnL while in position is computed as:

$$\text{PnL}_t = \left(P_{t-1}^{\text{straddle}} - P_t^{\text{straddle}}\right) \times \frac{\mathcal{N}}{S_{\text{entry}}} \tag{9}$$

where $P_t^{\text{straddle}} = C_t^{\text{mid}} + P_t^{\text{mid}}$ is the straddle mid-price, $\mathcal{N} = \$100{,}000$ is the notional, and S_{entry} is the ETF spot price at trade initiation. The contract count $\frac{\mathcal{N}}{S_{\text{entry}}}$ is fixed at entry to avoid implicit rebalancing.

Each position is delta hedged daily. The net delta of the short straddle is:

$$\Delta_{\text{net}} = -\left(\delta_{\text{call}} + \delta_{\text{put}}\right) \tag{10}$$

where δ_{call} and δ_{put} are sourced from the OptionMetrics delta field. The hedge quantity in shares is $h_t = \Delta_{\text{net}} \times (\mathcal{N}/S_{\text{entry}})$, and the daily hedge friction cost assumes one basis point of market impact per dollar of shares traded.

Transaction costs are applied at entry and exit as half the bid-ask spread on each option leg:

$$\begin{aligned}\mathrm{TC}_{\mathrm{entry/exit}} = &\left(\frac{O^C_{\mathrm{ask}} - O^C_{\mathrm{bid}}}{2} + \frac{O^P_{\mathrm{ask}} - O^P_{\mathrm{bid}}}{2}\right)_{\mathrm{ETF}} \times \frac{\mathcal{N}}{S_{\mathrm{entry}}} \\ &+ \sum_{i=1}^{N} w_i \left(\frac{O^C_{\mathrm{ask},i} - O^C_{\mathrm{bid},i}}{2} + \frac{O^P_{\mathrm{ask},i} - O^P_{\mathrm{bid},i}}{2}\right) \times \frac{\mathcal{N}}{S_{\mathrm{entry}}}\end{aligned} \tag{11}$$

This represents the cost of crossing from mid to the executable side of the market on both legs simultaneously. Positions are closed on the first date the exit condition is met.

3.6 Performance Evaluation

Performance metrics are computed by aggregating the position log produced by `strategy.py`. Net daily PnL is gross daily PnL minus the transaction cost incurred on that day, whether from entry, exit, or the delta hedge friction. Cumulative PnL series are constructed separately for each ETF as the running sum of daily net PnL over all dates where the position log records an active position.

The annualized Sharpe ratio is computed over all trading days in the position log, with zero PnL recorded on inactive days:

$$\mathrm{Sharpe} = \frac{\bar{r}_{\mathrm{net}}}{\hat{\sigma}_r} \times \sqrt{252} \tag{12}$$

where $\bar{r}_{\mathrm{net}}$ is the mean daily net PnL and $\hat{\sigma}_r$ is its standard deviation. Maximum drawdown is the largest peak-to-trough decline in the cumulative net PnL series. Regime-level Sharpe ratios are computed by splitting the daily PnL series at January 1, 2023, separating the 2022 high-volatility period from the 2023-2024 low-volatility period.

3.7 Volatility Surface Visualization

The volatility surface visualization is implemented in `src/visualizations/vol_surface.py`. For each ETF, the module identifies the date with the highest option chain density and plots all available option quotes on that date as a scatter plot in (log-moneyness, days-to-expiry, implied volatility) space. Because the 30-to-60-day expiry filter limits the available maturities to a handful of distinct expiration dates on any given trading day, the surfaces appear as a set of curves across strikes at each available maturity rather than a continuous mesh.

The resulting surfaces are shown in Figures 3a, 3b, and 3c: the smoothness of the SPY surface versus the roughness of the XBI surface reflect the underlying differences in option chain depth and data quality across the liquidity tiers.

4 Conclusions

4.1 Summary of Results

Table 1 presents the cross-ETF performance summary across the full sample period.

Table 1: Cross-ETF Backtest Summary (January 2022 – December 2024)

ETF	Tier	Trades	Avg Spread	Gross PnL	Txn Costs
SPY	1 (High)	17	0.006	\$3,518	\$2,114
XLK	2 (Medium)	16	0.013	\$2,246	\$11,794
XBI	3 (Low)	9	0.201	\$3,659	\$9,638

ETF	Net PnL	Sharpe (Net)	Max DD	Sharpe 2022 / 2023-24
SPY	\$1,403	0.11	\$5,746	−0.20 / 0.15
XLK	−\$9,548	−0.18	\$16,659	0.01 / −0.22
XBI	−\$5,978	−0.21	\$13,605	−0.25 / −0.20

4.2 The Implied Correlation Spread Confirms the Liquidity Hypothesis

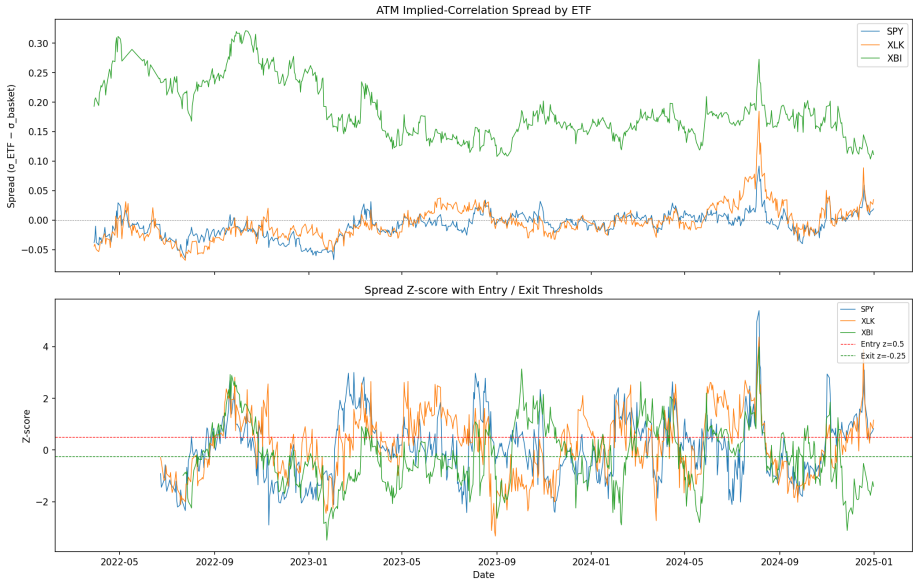


Figure 1: ATM Implied-Correlation Spread by ETF

The implied correlation spread levels strongly confirm the first part of the central empirical hypothesis. As shown in Figure 1, XBI sustained an average

spread of 18.5 volatility points throughout the sample, while SPY and XLK averaged −0.8 and −0.5 points respectively. The XBI spread was not only wider but structurally elevated for the entire 2022 period, spending much of the first half of the year between 0.20 and 0.31 before compressing modestly into 2023 and 2024. This is consistent with the hypothesis that less liquid ETF options markets allow a persistent wedge between the ETF's market-quoted implied volatility and the synthetic basket volatility derived from constituent options, because fewer arbitrageurs are positioned to close the gap.

SPY and XLK, by contrast, oscillated tightly around zero throughout the sample, frequently going negative, which indicates that in highly liquid markets the implied correlation spread is efficiently mean-reverting with no structural directional bias.

4.3 The Return Hypothesis Does Not Hold: Liquidity Improves Net Performance

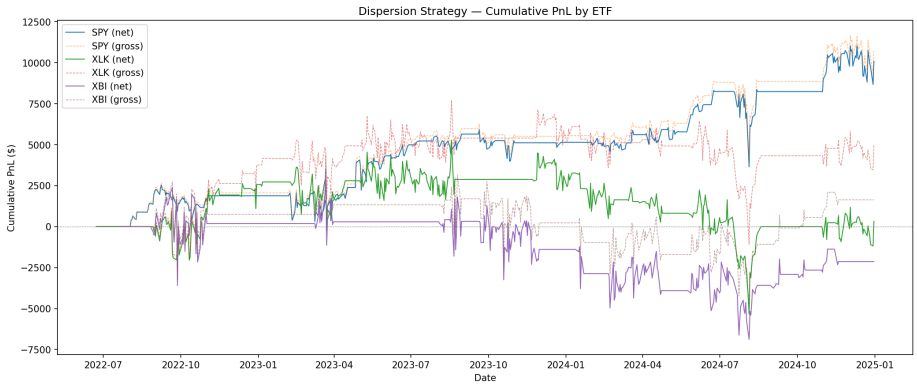


Figure 2: Cumulative PnL

Despite the spread level evidence, the second part of the hypothesis, that net strategy returns increase with decreasing liquidity up to a cost-dominated threshold, is decisively rejected. SPY was the only ETF with positive net PnL ($1,403), recording an annualized Sharpe ratio of 0.11 and a maximum drawdown of $5,746. XLK and XBI both produced net losses. The performance ordering across all three ETFs is monotonically increasing in liquidity, the opposite of what the hypothesis predicted.

The mechanism is transparent in the data. All three ETFs generated positive gross PnL before transaction costs: SPY at $3,518, XLK at $2,246, and XBI at $3,659, confirming that the spread signal identifies real mean-reversion opportunities across all liquidity tiers. However, the constituent straddle legs impose substantial per-name transaction costs at entry and exit. SPY paid $2,114 in total costs, a cost ratio of 60%. XLK paid $11,794 in costs, exceeding its gross PnL fivefold and producing a net loss of $9,548. XBI paid $9,638, producing a

net loss of $5,978. The wide bid-ask spreads on constituent single-name options, not the ETF leg, are the dominant cost driver: crossing 50 individual straddles for SPY or 20 biotech names for XBI on each round trip generates friction that overwhelms the gross edge in all but the most liquid tier.

4.4 Regime Decomposition

The regime-level Sharpe ratios reveal a mixed pattern. SPY recorded a Sharpe of -0.20 in 2022 and 0.15 in 2023-2024, performing better in the low-volatility regime. XLK was roughly flat in 2022 (0.01) and negative in 2023-2024 (-0.22). XBI was negative in both periods (-0.25 and -0.20).

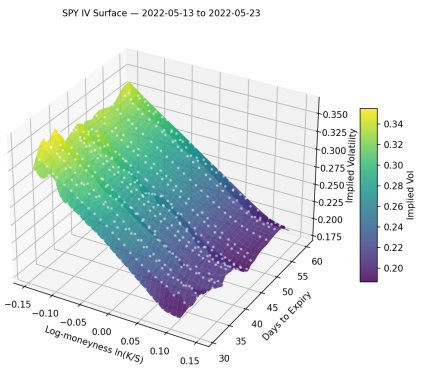
The reversal for SPY is driven by the constituent long straddles: during the high-volatility 2022 period, constituent option prices were elevated and volatile, and the mark-to-market losses on the long leg offset gains on the short ETF straddle. In the calmer 2023-2024 period, constituent straddle prices were more stable, allowing the short ETF leg's edge to flow through to net PnL. XLK and XBI remained unprofitable in both regimes, as constituent transaction costs dominated regardless of volatility conditions.

4.5 Volatility Surface Interpretation

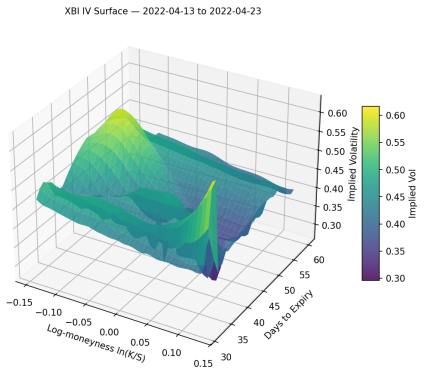
The implied volatility surfaces shown in Figures 3a, 3b, and 3c illustrate the qualitative differences between ETFs. The SPY surface exhibits a clean downward skew as log-moneyness moves from positive to negative, with data points densely distributed along each available maturity curve. The surface is densely populated with data points across the full log-moneyness band, reflecting deep and continuous liquidity. The XBI surface is considerably rougher, with visible discontinuities and localized spikes, reflecting gaps in the options market and stale quotes at strikes away from the money. The XLK surface occupies a middle ground, smoother than XBI but with a notable distortion near positive log-moneyness at shorter maturities. These surface characteristics reinforce why transaction costs and signal noise are more problematic for XBI and XLK: the raw implied volatility data contains more estimation error, which propagates into the implied correlation spread and produces noisier signals.

4.6 Limitations

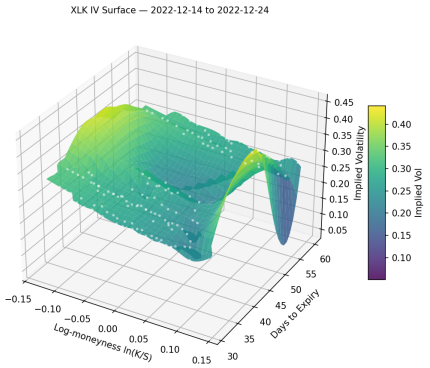
Several limitations are worth acknowledging. First, the backtest covers three ETFs over a single three-year window. The outperformance of SPY relative to XLK and XBI may partly reflect idiosyncratic characteristics of the 2022-2024 period rather than a general structural feature of the strategy. A longer sample covering multiple volatility cycles would strengthen the inference. Second, the constituent weight snapshots are averaged across three annual observations and held static within each year. For XBI, which has higher constituent turnover than the other two ETFs, this approximation introduces some error in the basket



(a) SPY IV Surface (May 2022)



(b) XBI IV Surface (April 2022)



(c) XLK IV Surface (December 2022)

Figure 3: Implied volatility surfaces across all three ETF liquidity tiers.

variance calculation, particularly around periods of significant index reconstitution. Third, the delta hedge is executed at daily frequency using closing deltas from OptionMetrics. Intraday delta drift and the inability to hedge continuously means the backtest overstates the ease of maintaining a delta-neutral position, especially during high-volatility episodes. Fourth, the constituent options used to construct the synthetic basket vol are themselves filtered to the same 30-to-60-day maturity window and ATM band as the ETF options, which means the basket vol estimate is not always computed from a complete constituent set on days where individual names lack liquid near-term options. Fifth, constituent implied volatilities that are missing on isolated dates are forward-filled before computing the basket variance, which introduces stale-quote bias on precisely the days when option markets are thin or stressed.

4.7 Directions for Further Work

The most natural extension of this work is to include a wider cross-section of ETFs spanning a broader range of liquidity tiers, to test whether the monotone relationship between liquidity and net returns observed here generalizes. A transaction-cost-aware signal threshold, where entry is only triggered when the spread is sufficiently wide to cover expected round-trip costs at the prevailing bid-ask spread, could also recover profitability in the medium and low-liquidity tiers that the fixed z-score threshold fails to achieve. Finally, allowing the constituent weight vector to update continuously using SEC N-PORT filings rather than annual snapshots would reduce holdings approximation error and make the basket variance calculation more precise throughout the backtest window.